

THIS IMAGE IS UNDER INVESTIGATION.

SAFAID asks every image the same forensic question, with no category labels, and tells real photographs from AI-generated images across 19 generators with one fine-tuned 4B vision–language model.

89.52%	0	1
AVERAGE ACCURACY ON 19 GENERATORS ± 1.01 OVER 3 RUNS · +1.70 OVER CATAID	CATEGORY LABELS IN TRAINING AND AT TEST TIME	GPU, 4.44 B PARAMETERS CATAID: 9 B PARAMETERS, 8 GPUS

HOW IT WORKS

01

Ask one question

Every image gets the same prompt, in training and at test time. The content of the image is never named.

“This is an image that is under investigation. Is this image generated by AI, or is it a real image?”

02

Augment only the fakes

Each generated training image gets three degraded copies. Real images stay untouched, as they are in the wild.

real

fake + 3 degraded copies

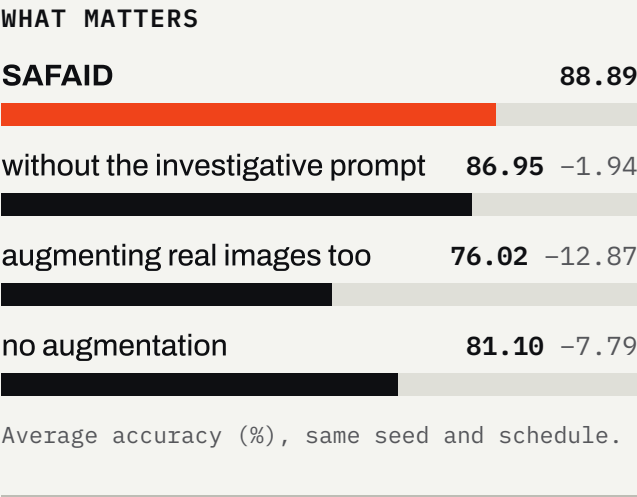
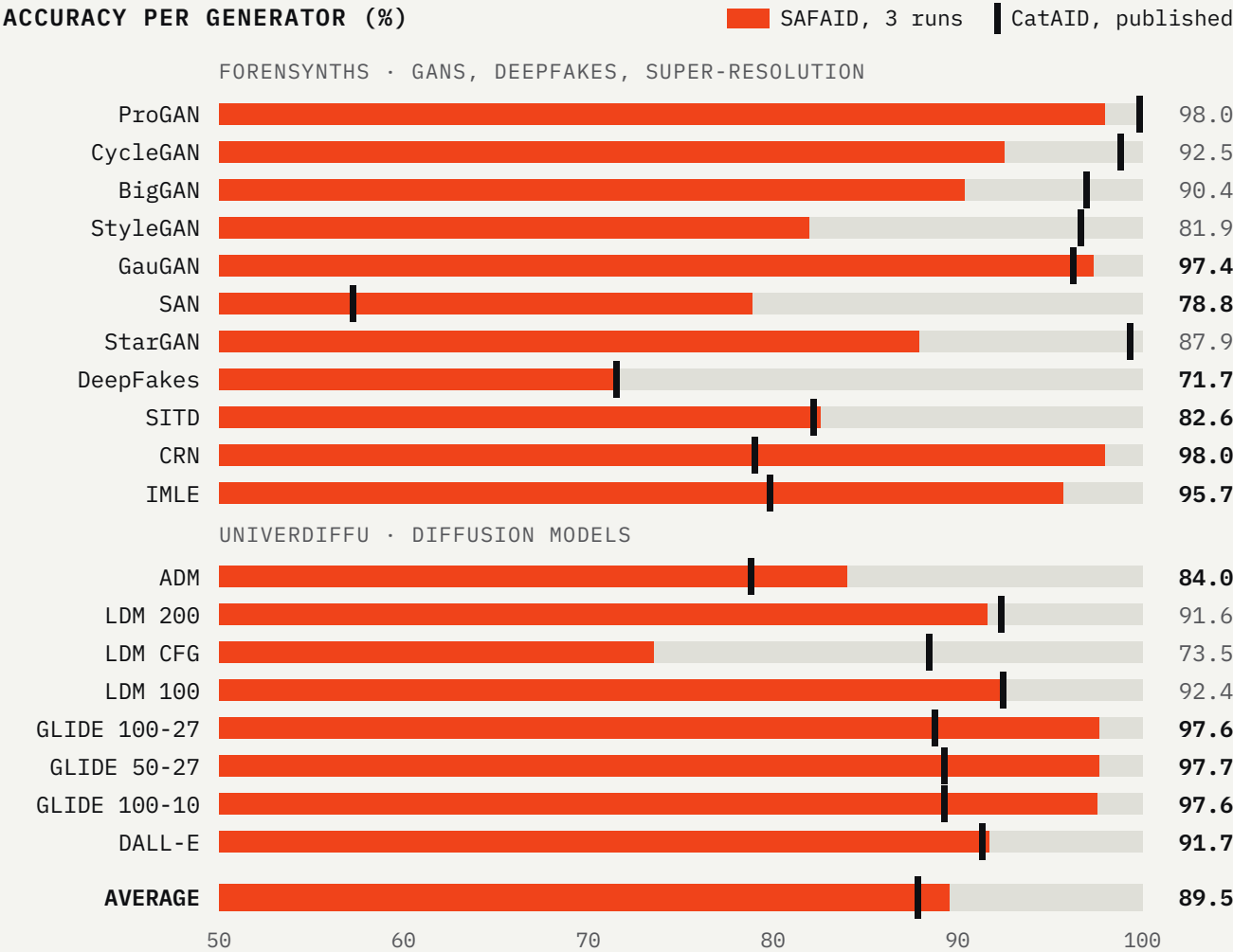
03

Train one compact model

All 4.44 B parameters of Qwen3-VL-4B are fine-tuned on 10,000 ProGAN-based samples. The answer is one of two sentences.

This is a real image.

This is an AI-generated image.



TAKEAWAYS

Category labels are not needed. One context-free question is enough.

Augment only the fakes. Degrading real images too is worse than no augmentation.

Mind the asymmetry. 96.4% of real but 82.6% of generated images are classified correctly: a “fake” verdict is the stronger evidence.



CODE, DATA MANIFESTS AND ALL RUN RECORDS
github.com/mirrorcall196/SAFAID

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